Project : Linux User and File Permission Management

Project Description

This project focuses on managing users, groups, and file permissions in a Linux environment. It demonstrates how to control access to files and directories using Linux commands to ensure system security

Skills Gained

- managing users and group permissions
- Managing directory permission
- managing file permission
- control access to files
- control access to directories

1. Types of Files

1. **User Defined Files:** User Defined Files are created by user which are human readable
   Types of User Defined Files:
   1. Normal File : It create by touch vim cat nano
   2. Directory File : It create by mkdir
   3. Link File : create by ln
      There are two types of link file:
      - Hard Link : Link count 2 or more and they look like normal which first character in permission string is _
      - Soft Link: Link count 1 and the first character in the permissions string is l
2. **System Defined Files:** System Defined file are created by system
   Types of System Defined Files:
   1. Character File : The first character in the permissions string is c
   2. Block files: The first character in the permissions string is b
   3. Socket file: The first character in the permissions string is s
   4. Pipe file: The first character in the permissions string is p

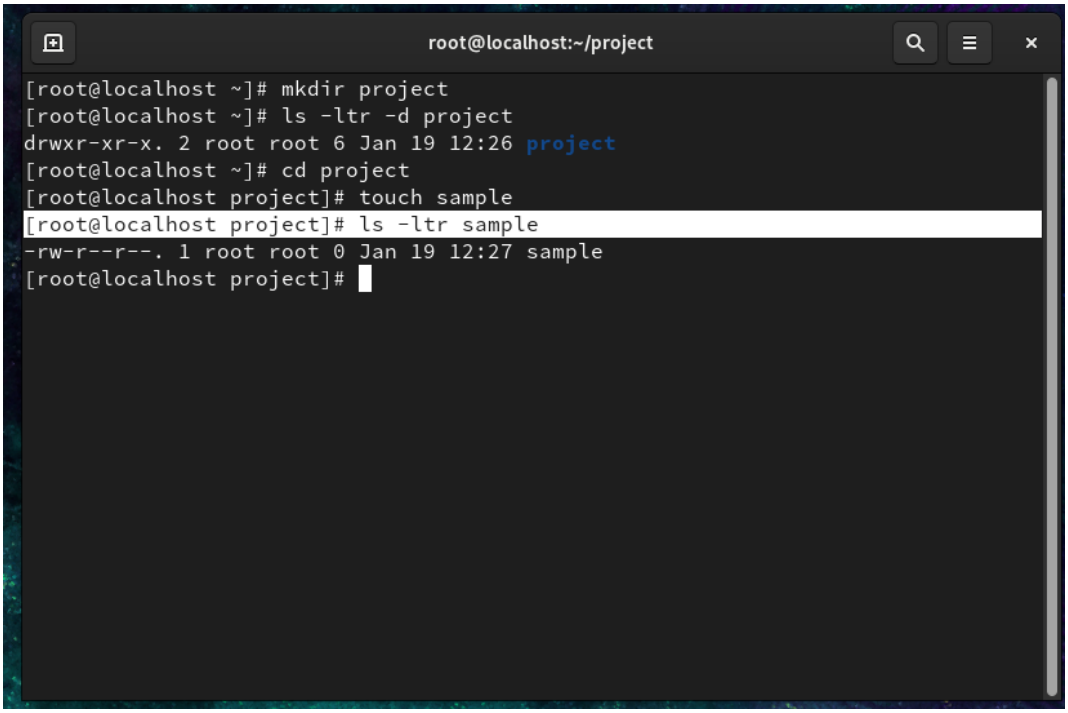
2. Types of Permission

1. Owner Permission

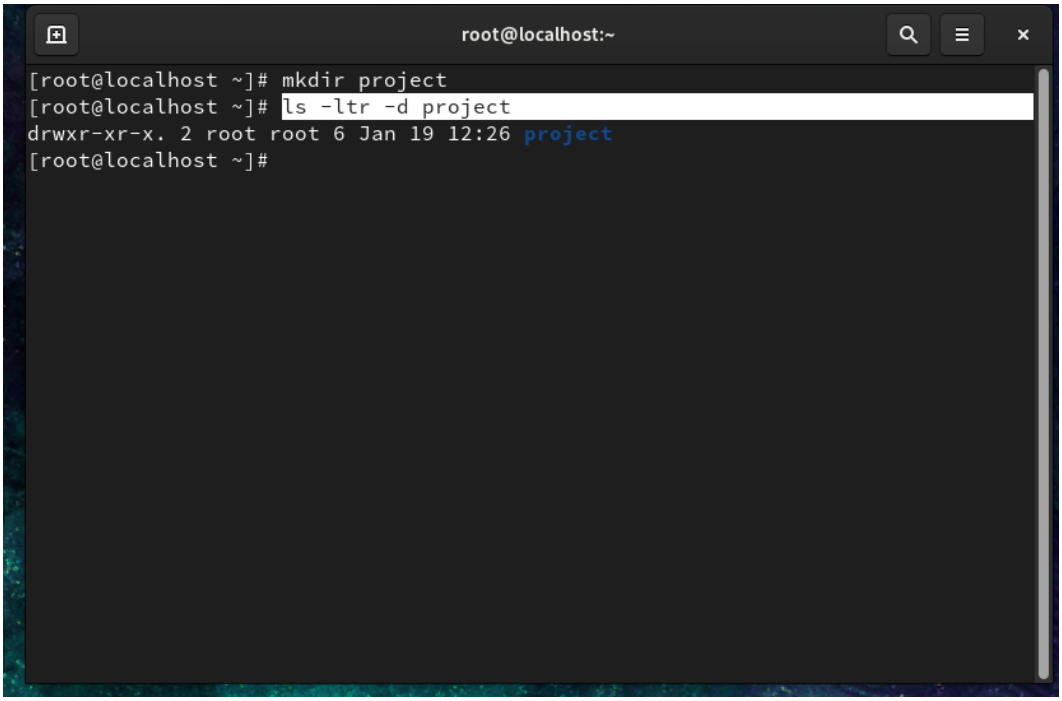
2. Group Permission

3. Others Permission

- ls -ltr sample : It use for see the permission of file



- Ls –ltr –d project: It use for see the permission of directory



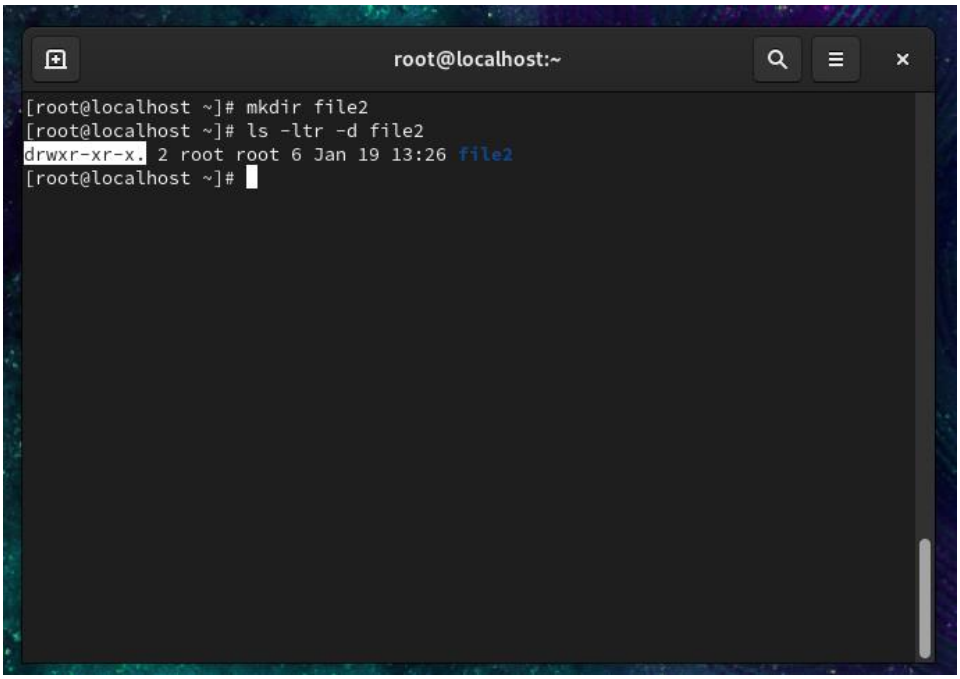
- **Hardlink** has link count 2 and it looks like normal

```
root@localhost:~
[root@localhost ~]# touch original.txt
[root@localhost ~]# ln original.txt hardlink.txt
[root@localhost ~]# ln -s original.txt hardlink.txt
ln: failed to create symbolic link 'hardlink.txt': File exists
[root@localhost ~]# ln -s original.txt softlink.txt
[root@localhost ~]# ls -l
total 8
-rw-------. 1 root root 1104 Dec 31 22:23 anaconda-ks.cfg
-rw-r--r--. 1 root root    0 Jan 19 13:24 demo
-rw-r--r--. 1 root root    6 Jan 19 12:39 demo1
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Desktop
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Documents
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Downloads
drwxr-xr-x. 2 root root    6 Jan 19 13:26 file2
-rw-r--r--. 2 root root    0 Jan 19 13:46 hardlink.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Music
-rw-r--r--. 2 root root    0 Jan 19 13:46 original.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Pictures
drwxr-xr-x. 4 root root   85 Jan 19 12:36 project
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Public
lrwxrwxrwx. 1 root root   12 Jan 19 13:46 softlink.txt -> original.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Templates
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Videos
[root@localhost ~]#
```

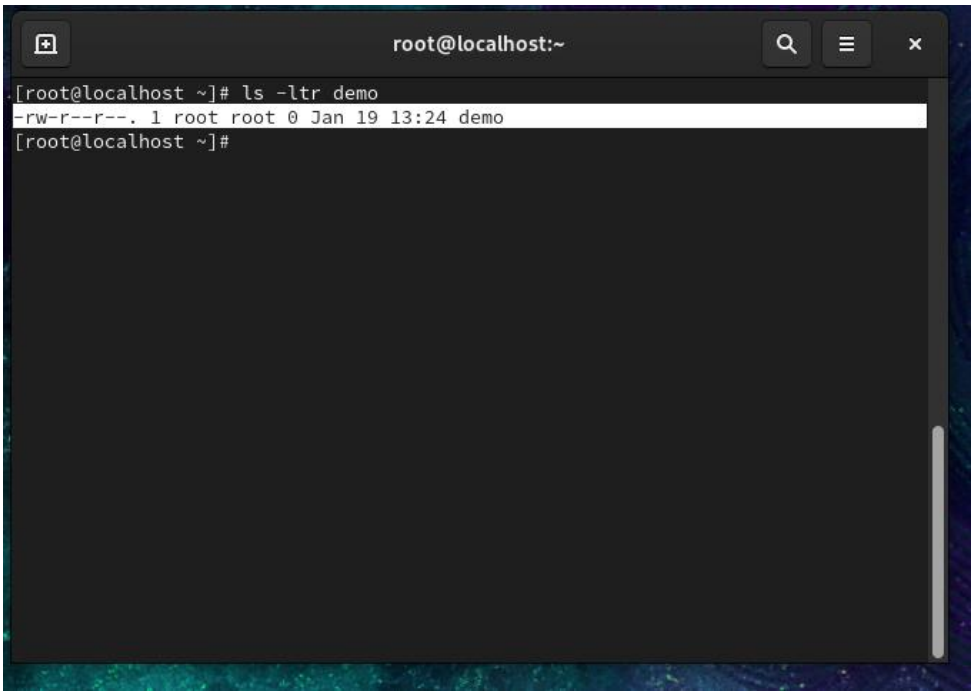
- **Softlink** has link count 1 and it start with lr

```
root@localhost:~
[root@localhost ~]# touch original.txt
[root@localhost ~]# ln original.txt hardlink.txt
[root@localhost ~]# ln -s original.txt hardlink.txt
ln: failed to create symbolic link 'hardlink.txt': File exists
[root@localhost ~]# ln -s original.txt softlink.txt
[root@localhost ~]# ls -l
total 8
-rw-------. 1 root root 1104 Dec 31 22:23 anaconda-ks.cfg
-rw-r--r--. 1 root root    0 Jan 19 13:24 demo
-rw-r--r--. 1 root root    6 Jan 19 12:39 demo1
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Desktop
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Documents
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Downloads
drwxr-xr-x. 2 root root    6 Jan 19 13:26 file2
-rw-r--r--. 2 root root    0 Jan 19 13:46 hardlink.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Music
-rw-r--r--. 2 root root    0 Jan 19 13:46 original.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Pictures
drwxr-xr-x. 4 root root   85 Jan 19 12:36 project
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Public
lrwxrwxrwx. 1 root root   12 Jan 19 13:46 softlink.txt -> original.txt
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Templates
drwxr-xr-x. 2 root root    6 Jan 10 10:53 Videos
[root@localhost ~]#
```

- Directory file string with d



- Normal Files strating with _



- **Link Count** is incresing when we perform some action in directory

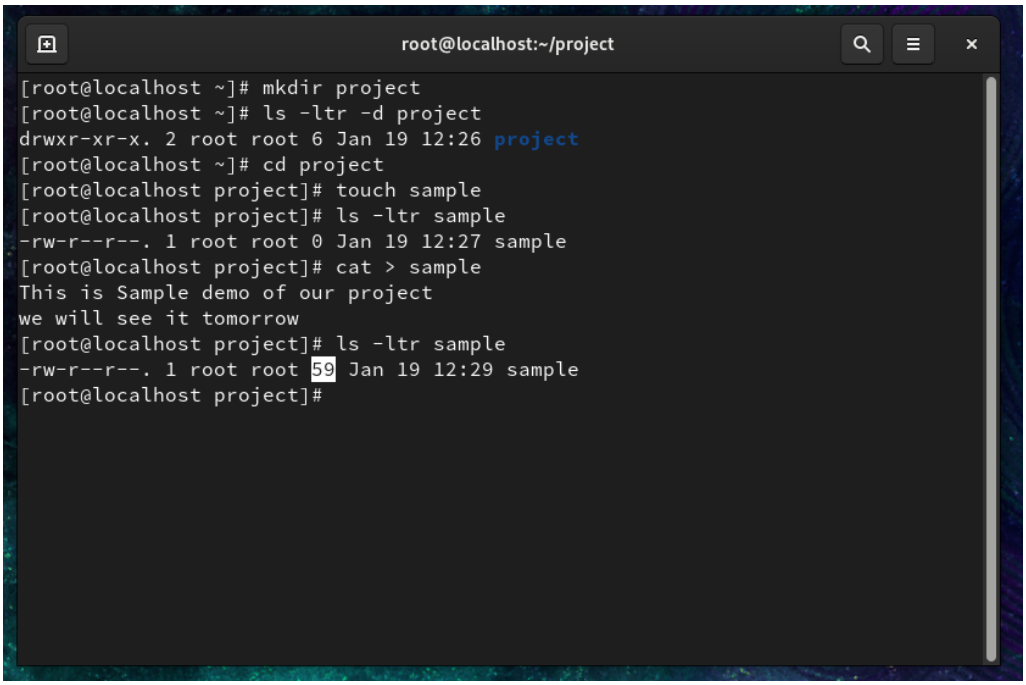
For Example: When we add directory in directory, So link count is increase

```
root@localhost:~
[root@localhost ~]# cd project
[root@localhost project]# mkdir demo1
[root@localhost project]# touch file1
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 3 root root 46 Jan 19 12:33 project
[root@localhost ~]# cd project
[root@localhost project]# mkdir demo2
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 4 root root 59 Jan 19 12:34 project
[root@localhost ~]# cd project
[root@localhost project]# touch file2
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 4 root root 72 Jan 19 12:35 project
[root@localhost ~]# cd project
[root@localhost project]# cat > file3
file3
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 4 root root 85 Jan 19 12:36 project
[root@localhost ~]#
```

- when we add file in directory , link count doesn't increase

```
root@localhost:~
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 2 root root 20 Jan 19 12:27 project
[root@localhost ~]# cd project
[root@localhost project]# mkdir demo1
[root@localhost project]# touch file1
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 3 root root 46 Jan 19 12:33 project
[root@localhost ~]# cd project
[root@localhost project]# mkdir demo2
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 4 root root 59 Jan 19 12:34 project
[root@localhost ~]# cd project
[root@localhost project]# touch file2
[root@localhost project]# cd ..
[root@localhost ~]# ls -ltr -d project
drwxr-xr-x. 4 root root 72 Jan 19 12:35 project
[root@localhost ~]#
```

- **5th field** is indicated as size of file



NAME : VAISHNAVI MULE